

OVARIAN CANCER METASTASIS TO THE BREAST: A DIAGNOSTIC PITFALL MIMICKING A PROBABLY BENIGN BREAST LESION

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INTRODUCTION

Breast metastasis from extramammary malignancies is a rare occurrence, accounting for 0,17% of metastases from gynecological primary tumor, with ovarian malignancies metastasis occurrence of only 0,07% of cases.^{1,2} Metastatic lesions may present similarly with primary breast tumor and associated with poor prognosis of the primary malignancy.³ Failure to recognize a metastatic origin may result in incorrect staging, unnecessary breast-directed surgery, and inappropriate treatment.

CASE PRESENTATION

A 43-year-old female with chief complaint of abdominal distension and suspected ovarian malignancy from prior hospital was referred to our hospital for surgery and chemotherapy. Preoperative CT showed multiloculated cyst with enhanced solid component from the ovary, suggestive of malignancy. Abdominal hysterectomy with bilateral salpingo-oophorectomy was performed followed by chemotherapy.

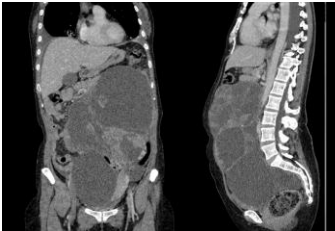


Figure 1. CT shows multiloculated cyst with solid component from ovary.

On interim CT, a solid right breast lesion was found. In retrospect, subtle solid breast lesion can be found in previous CT, and on follow-up CT after two months, significant growth and a conspicuous lesion is observed.

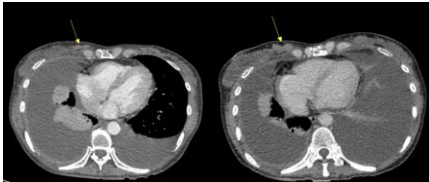


Figure 2. CT shows solid lesion with significant growth within two month.

No pain or tenderness noted, breast ultrasound was done for further evaluation, and it shows well-circumscribed hypoechoic lesion. US-guided biopsy was performed.

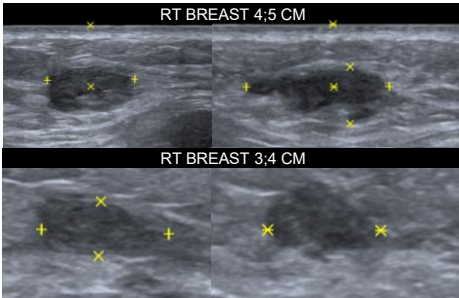


Figure 3. US shows parallel ovoid hypoechoic, well-circumscribed lesion.

Histopathology results show evidence for adenocarcinoma, favoring metastasis. No immunohistochemistry examination was performed.

DISCUSSION

Breast metastasis lesion may have atypical appearances. In this case, a well-circumscribed hypoechoic lesion could make a false judgement favoring benign breast lesion such as fibroadenoma. Ovarian cancer that metastasizes to the breast is hard to distinguish from primary breast cancer due to similarities in morphology and immunohistochemical profile. Integration of the patient's oncologic history, clinical context, imaging findings, and pathological assessment is essential in establishing the true pathology. Histopathology shows adenocarcinoma that aligns with metastasis in this patient.

Breast metastasis could give out a benign-appearing imaging feature, in patients with extramammary primary malignancy, benign-like lesion should be carefully considered for higher staging of BI-RADS and be considered higher in differential diagnosis rather than favoring only on morphological assessment. A multidisciplinary approach integrating clinical, radiological, and pathological findings is crucial for accurate diagnosis. Histopathology and immunochemistry tests should be done with proper and careful clinical correlation to ensure appropriate treatment for the patient.

CONCLUSION

Breast metastases may mimic probably benign lesions, highlighting the importance of multidisciplinary approach for accurate diagnosis.

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